COMPUTATIONAL PROTEOMICS · 2026

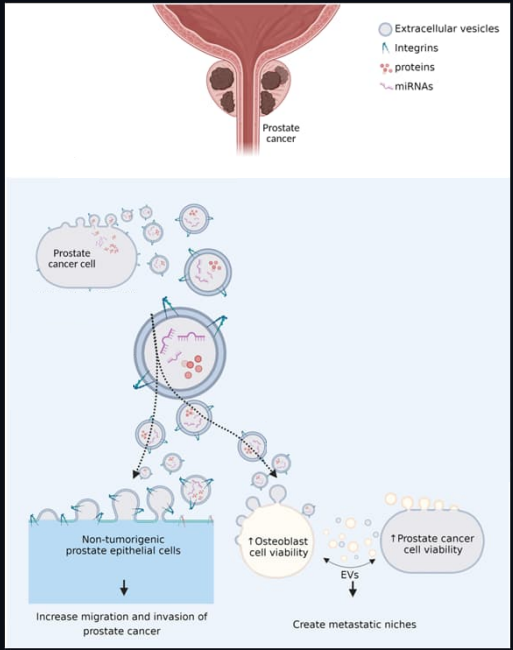
SILAC-TMT-RTS-SPS-MS3 Pipeline

Quantitative Proteomic Profiling of Differential Extracellular Vesicle Uptake in Osteoblastic Niches

Resolving Selective EV Internalization via Multiplexed Isobaric Labeling, Robust Scaling, and Orthogonal Machine Learning

Work done and presented by Vladislav Mantsevich

October 22, 2026



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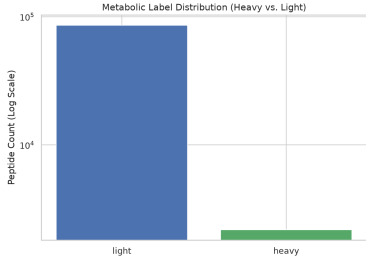
Discrimination of EV Cargo: SILAC-TMTpro 18-plex Workflow

- **Pulse-Chase Tracing**: Donor PC3/PC3M cells pulse-labeled with heavy Arg10/Lys-D8; recipient SAOS-2 acts as unlabelled light background (~97.5% total signal).
- **Sample Layout (TMTpro 18-plex):** 7 uptake conditions (PC3 N=3, PC3M N=4), 3 unlabelled SAOS-2 baselines, negative controls, and carrier/boost channels.
- **SPS-MS3 Protocol**: Orbitrap Eclipse with synchronous precursor selection (SPS-MS3) and co-isolation threshold <75% to eliminate ratio compression.
- **Stringent ID**: Sequest HT / Percolator target-decoy scoring (q <= 0.01 at PSM and protein level).

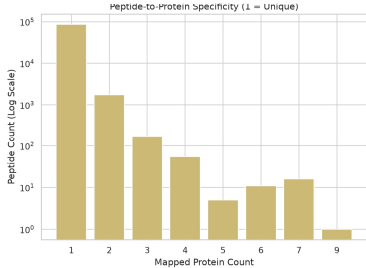
TMT channel	Sample content
126	PC3 Evs
127N	PC3M Evs
127C	
128N	
128C	SAOS2
129N	SAOS2
129C	SAOS2
130N	SAOS2 + PC3 EV
130C	SAOS2 + PC3 EV
131N	SAOS2 + PC3 EV
131C	SAOS2 + PC3M EV
132N	SAOS2 + PC3M EV
132C	SAOS2 + PC3M EV
133N	SAOS2 + spike PC3 EV
133C	SAOS2 + spike PC3 EV
134N	SAOS2 + spike PC3M EV
134C	SAOS2 + spike PC3M EV
135N	SAOS2 + PC3M EV

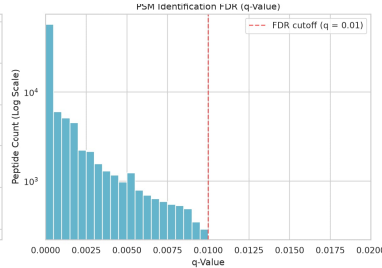
Upstream Quality Control (MS1 and MS2 Levels)

1. **Metabolic Label Ratio**: Proportion of Heavy to Light peptides.
2. **Contaminant Annotation**: Verification that external contaminants were filtered out.
3. **Peptide Uniqueness**: Frequency of peptide matches mapping to a distinct proteins.
4. **PSM False Discovery Rate**: Global q-value distribution to ensure high-confidence identification thresholds.



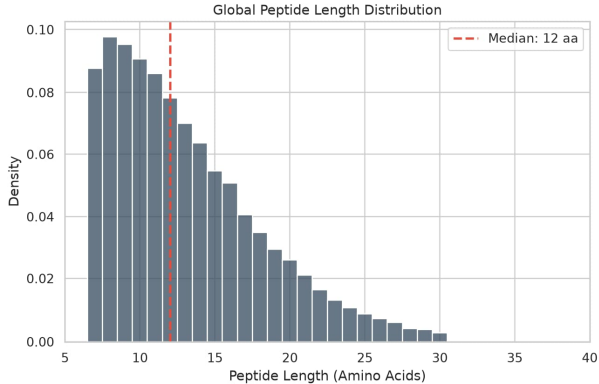


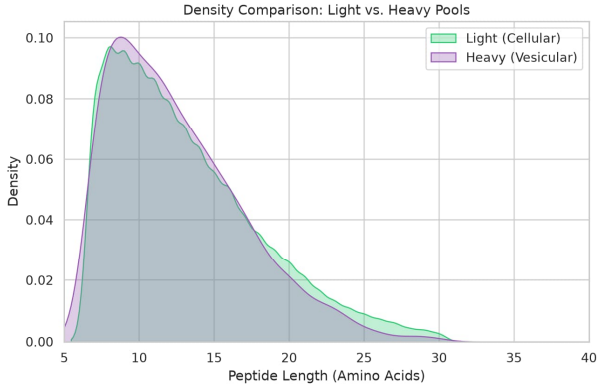




Enzymatic Cleavage and Peptide Length Profiling

```
Sequence length summary metrics:
Total peptides: 88138
Minimum length: 7 aa
Median length: 12.0 aa
Mean length: 13.13 aa
Maximum length: 30 aa
```

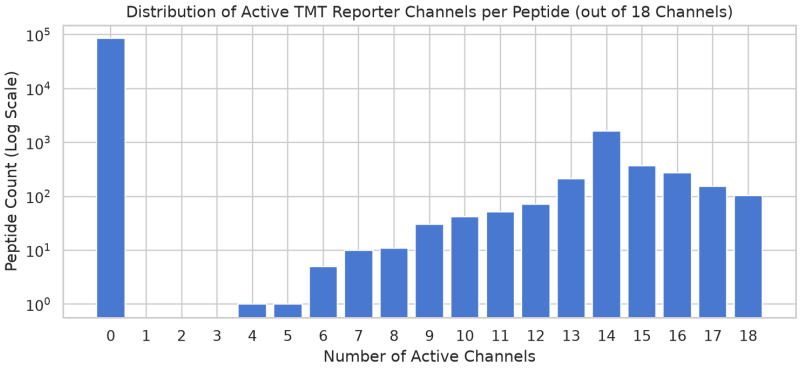




TMT Multiplex Reporter Occupancy Diagnostics

Channel summary by isotopic label:			
Label	count	median	mean
heavy	2209	0.0	6.63
light	85929	0.0	0.32

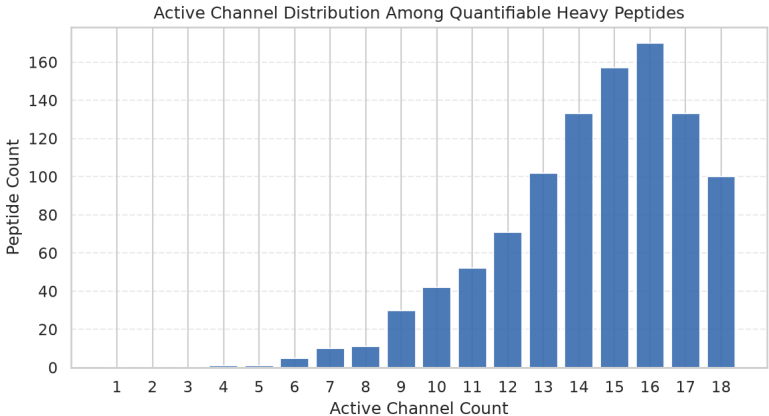
Within the target heavy subset (n=2,209), more than a half contain zero reporter intensities across all channels. Why?




```
Quantification status for Heavy peptides:
Quan Info
NoQuanValues        1175
NaN                 1018
ExcludedByMethod      16
Name: count, dtype: int64
```

Quantifiable Heavy Peptide Cohort

We isolate the 1,018 quantifiable heavy peptides (where are at least 1 non-zero channel) and examine their channel occupancy across replicates. 7 of them have signal in empty and/or spike channels, and not in uptake channels – remove them

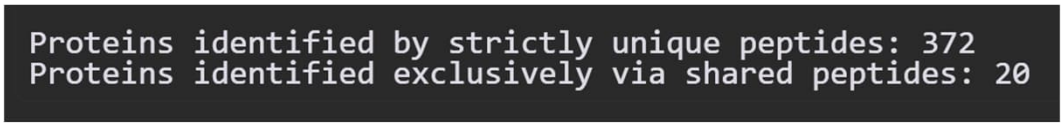


Protein Group Resolution and Shared Peptide Evaluation

All 1,018 peptides belong to a single protein group (Number of Protein Groups == 1).

However, 73 peptides map to multiple closely related proteins within the group (Number of Proteins > 1).

Strict exclusion of these 73 shared peptides would completely discard 20 proteins, including canonical EV cytoskeletal markers (ACTB, ACTC1, TUBA4A, VIM) whose tryptic fragments are identical between isoforms. Therefore, we track both cohorts: derived exclusively from 945 strictly unique peptides (372 proteins) (done), and derived from all 1,011 peptides (392 proteins) (in progress)



One sample has 3.5 times less cumulative intensity than others – outlier!

Get rid of him

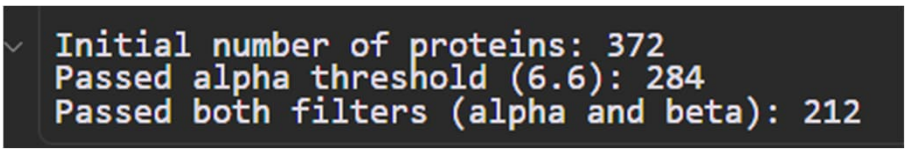
	Total_Intensity	Median_Intensity	Quantified_Proteins	Zero_Count
PC3_EV_R1	2266.9	2.90	249	123
PC3_EV_R2	7448.2	9.00	346	26
PC3_EV_R3	8614.1	10.70	351	21
PC3M_EV_R1	7073.7	10.60	343	29
PC3M_EV_R2	9511.2	13.15	364	8
PC3M_EV_R3	10844.9	15.25	370	2
PC3M_EV_R4	7100.4	10.55	357	15

Background Noise Filtering Against Recipient Cell Controls

To eliminate non-specific background signals, a filter is being applied relative to the SAOS-2 alone control channels:

1. Peak sample median intensity must be higher than noise:
   Max[median(PC3 + EV), median(PC3M + EV)] > 6.6

1. Sample signal must be at least twofold above background:
   *Max[median(PC3 + EV), median(PC3M + EV)] >= 2 * median(SAOS2 control)*



Spike-in Normalization and Input Scaling

To account for EV input loading differences, sample intensities are normalized to the internal pure-EV spike channels.

A 2x NTA particle adjustment factor is applied to PC3 channels (−1.0 on the log2 scale):

$$\log_2(I_{norm}) = \log_2(I_{sample}) - \log_2(I_{spike}) - offset$$

Missing values (zeros) are assigned empirical LOD/2 values

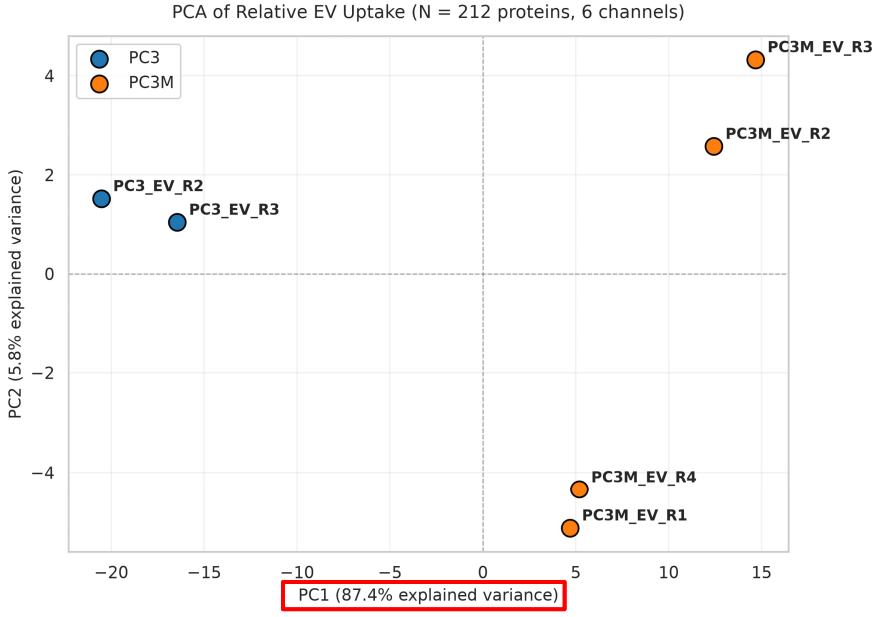
```
=== Missing values and zeros by channel ===
              Zero_Count  NaN_Count  Total_Missing
PC3_EV_R2              1          0              1
PC3_EV_R3              0          0              0
PC3M_EV_R1             1          0              1
PC3M_EV_R2             0          0              0
PC3M_EV_R3             0          0              0
PC3M_EV_R4             0          0              0
spike_PC3_R1           0          0              0
spike_PC3_R2           0          0              0
spike_PC3M_R1          0          0              0
spike_PC3M_R2          0          0              0

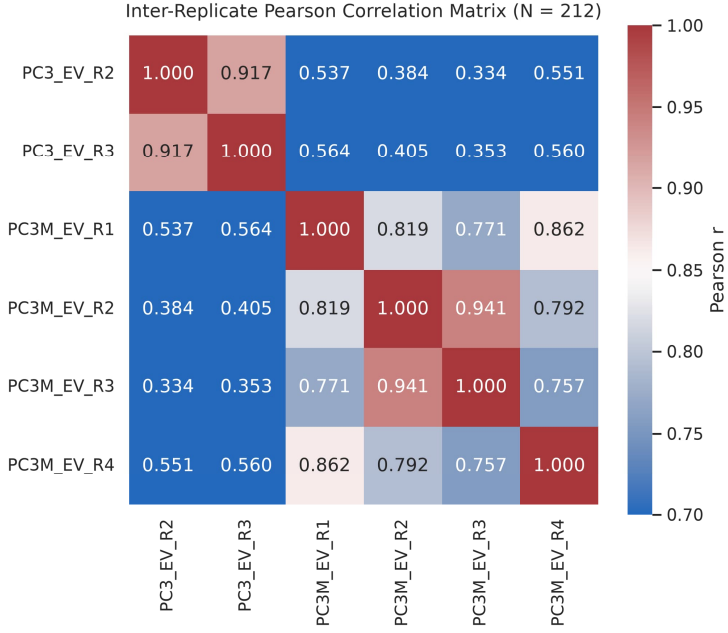
=== Distribution of missing values across proteins (among 6 experimental channels) ===
0    210
1      2
Name: count, dtype: int64

Proteins with no missing values (zeros) across all 6 channels: 210 out of 212
```

EV-cell incubation

To prevent excessive differences in quantities of isolated PC3 and PC3M EVs, their concentration was determined by NTA. Briefly, 1 µl of resuspended EVs was added to 999 µl PBS and measured on 3 different camera levels (12, 14, 16) for 3 times 30 seconds. The average of these measurements showed that there were ~2× more PC3 EVs than PC3M EVs. Since we did not want to risk adding a lower amount of PC3 EVs than with our initial EV*trace* experiments (REF), we decided to add an equal volume of the different EVs to the cells, even if this would mean that we added twice as much PC3 EVs than PC3M EVs. Thus in all cases, the amount of added EVs was derived from one T175 flask, which was equal to 30 mL of EV collection medium. The 58 µl of resuspended PC3 or PC3M EVs was mixed with 100 µl of McCoy 5a Glutamax medium (Gibco) supplemented with 2% FBS and pen/strep. Seeded SAOS2 cells in the 48 wells plate were first washed twice with 500 µl PBS before adding 150 µl of McCoy 5a + 2%


PCA of Relative EV Uptake (N = 212 proteins, 6 channels)
PC3
PC3M
PC3M_EV_R3
PC3M_EV_R2
PC3_EV_R2
PC3_EV_R3
PC3M_EV_R4
PC3M_EV_R1
PC2 (5.8% explained variance)
PC1 (87.4% explained variance)
4
2
0
−2
−4
−20
−15
−10
−5
0
5
10
15


Inter-Replicate Pearson Correlation Matrix (N = 212)
PC3_EV_R2
PC3_EV_R3
PC3M_EV_R1
PC3M_EV_R2
PC3M_EV_R3
PC3M_EV_R4
1.000 0.917 0.537 0.384 0.334 0.551
0.917 1.000 0.564 0.405 0.353 0.560
0.537 0.564 1.000 0.819 0.771 0.862
0.384 0.405 0.819 1.000 0.941 0.792
0.334 0.353 0.771 0.941 1.000 0.757
0.551 0.560 0.862 0.792 0.757 1.000
PC3_EV_R2
PC3_EV_R3
PC3M_EV_R1
PC3M_EV_R2
PC3M_EV_R3
PC3M_EV_R4
Pearson r
1.00
0.95
0.90
0.85
0.80
0.75
0.70

Identification of the Global Scale Effect

(almost) all the PC1 feature coefficients are positive!!! (from -0.006 to +0.073)

In uncentered PCA, a component with uniform positive signs reflects a **global scale factor**

Meaning, recipient SAOS-2 cells internalize metastatic PC3M cargo at a systematically higher absolute volume than PC3 cargo.

```
PC1 Loadings: min = -0.0628, max = +0.0734
Proportion of positive loadings: 211 / 212
```

Sample-Median Centering: Isolating Relative Cargo Composition

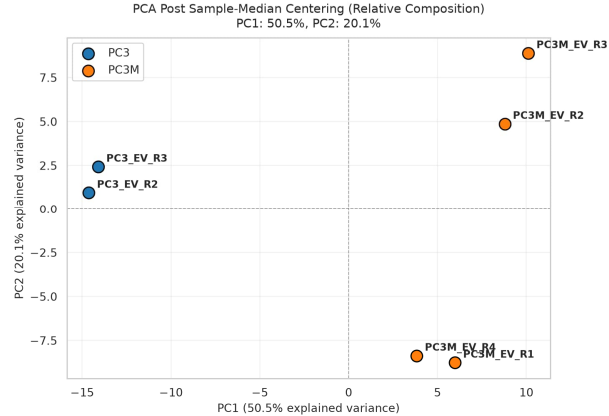
To isolate specific cargo selectivity from bulk endocytosis rates, we subtract sample medians:

$$Signal_{rel;\,peptide,sample} = Signal_{peptide,sample} - Median(Signal_{peptide,sample})_{sample}$$

```
=== Top Proteins with Relative Cargo Selectivity (PC3M enriched) ===
     Genes  PC1_Loading_Centered  Relative_Delta_PC3M_vs_PC3
42  COL4A2                0.095764                6.014406
75    TGM2                0.094832                4.126660
23     FN1                0.094745                3.761799
78     TNC                0.091768                2.106766
125  ITGA1                0.093116                2.012386
13  CTNND1                0.096061                1.973426
195   EHD1                0.054657                1.815406
58  ANPEP                 0.095653                1.791282
4     CCN1                0.093940                1.705974
1    CLIC1                0.079106                1.623519

=== Top Proteins with Relative Cargo Selectivity (PC3 enriched) ===
     Genes  PC1_Loading_Centered  Relative_Delta_PC3M_vs_PC3
168  MAP3K12               -0.095923               -2.249427
186     HM13               -0.094925               -1.768839
169    LMAN2               -0.080420               -1.699602
192     GDF15              -0.093447               -1.468005
127  S100A10               -0.074820               -1.464022
57   HSP90B1               -0.093001               -1.298344
182    CD276               -0.082978               -1.258369
45      CD63               -0.091987               -1.245064
116  ALDH3A2               -0.082075               -1.228481
209    HYOU1               -0.071250               -1.216590
```

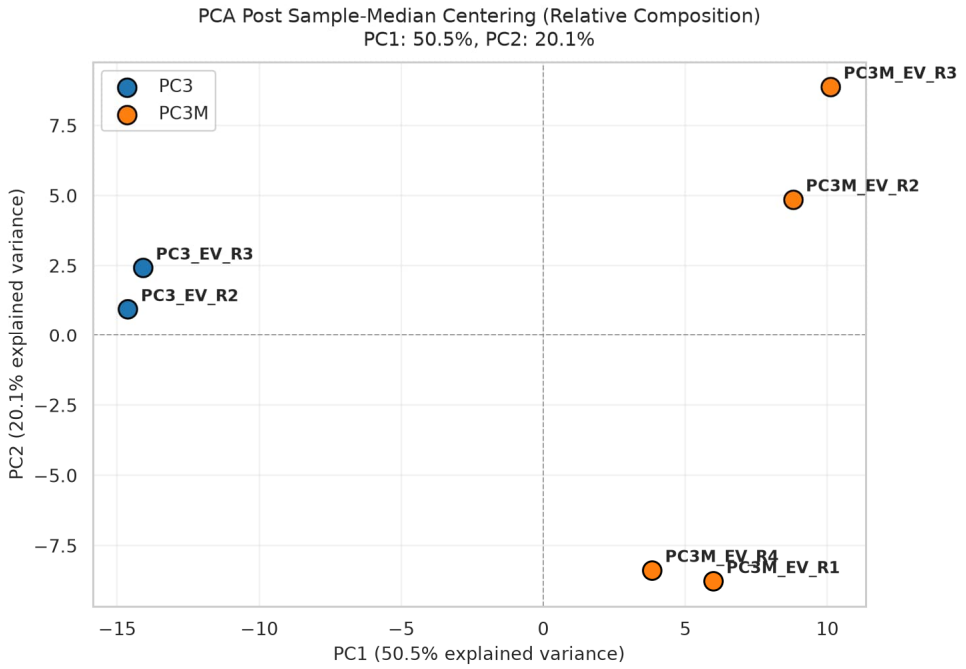
+ Code + Markdown



```
=== Top Proteins with Relative Cargo Selectivity (PC3M enriched) ===
      Genes  PC1_Loading_Centered  Relative_Delta_PC3M_vs_PC3
42   COL4A2              0.095764                   6.014406
75     TGM2              0.094832                   4.126660
23      FN1              0.094745                   3.761799
78      TNC              0.091768                   2.106766
125   ITGA1              0.093116                   2.012386
13   CTNND1              0.096061                   1.973426
195    EHD1              0.054657                   1.815406
58    ANPEP              0.095653                   1.791282
4      CCN1              0.093940                   1.705974
1     CLIC1              0.079106                   1.623519

=== Top Proteins with Relative Cargo Selectivity (PC3 enriched) ===
      Genes  PC1_Loading_Centered  Relative_Delta_PC3M_vs_PC3
168  MAP3K12             -0.095923                  -2.249427
186     HM13             -0.094925                  -1.768839
169    LMAN2             -0.080420                  -1.699602
192    GDF15             -0.093447                  -1.468005
127  S100A10             -0.074820                  -1.464022
57   HSP90B1             -0.093001                  -1.298344
182    CD276             -0.082978                  -1.258369
45      CD63             -0.091987                  -1.245064
116  ALDH3A2             -0.082075                  -1.228481
209    HYOU1             -0.071250                  -1.216590
```

+ Code + Markdown


PCA Post Sample-Median Centering (Relative Composition)
PC1: 50.5%, PC2: 20.1%
PC3
PC3M
PC3M_EV_R3
PC3M_EV_R2
PC3_EV_R3
PC3_EV_R2
PC3M_EV_R4
PC3M_EV_R1
PC2 (20.1% explained variance)
PC1 (50.5% explained variance)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−15
−10
−5
0
5
10

Univariate Differential Uptake Analysis: Welch's t-Test

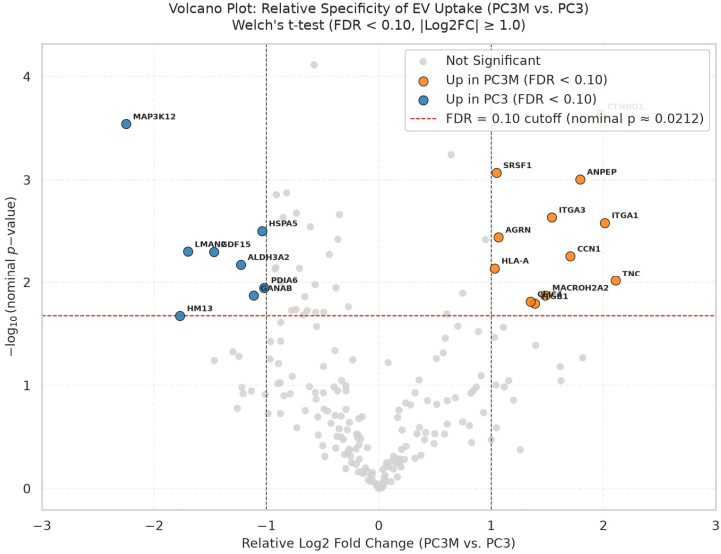
1. Heteroscedasticity:
   MS signal variance
   scales with intensity

2. Unbalanced Groups:
   n_1 = 2, n_2 = 4

Student's t-test assumptions are not met, but its variation – (two-sided) Welch's t-test – would work here well?

Multiple testing correction is applied using the Benjamini–Hochberg False Discovery Rate:

FDR < 0.10,
log_2(FC) >= 1.0



Univariate Differential Uptake Analysis: Welch's t-Test

Total proteins passing FDR < 0.10 and |Log2FC| >= 1.0: 22

Master Protein Accessions	Gene	Log2FC	p_value	q_value	Significance
O60716	CTNND1	1.973426	0.000230	0.015249	Up in PC3M (FDR < 0.10)
Q12852	MAP3K12	-2.249427	0.000288	0.015249	Up in PC3 (FDR < 0.10)
P15144	ANPEP	1.791282	0.000988	0.029911	Up in PC3M (FDR < 0.10)
Q07955	SRSF1	1.046582	0.000852	0.029911	Up in PC3M (FDR < 0.10)
P08572	COL4A2	6.014406	0.001556	0.032989	Up in PC3M (FDR < 0.10)
P26006	ITGA3	1.541898	0.002305	0.034981	Up in PC3M (FDR < 0.10)
P56199	ITGA1	2.012386	0.002600	0.036741	Up in PC3M (FDR < 0.10)
P11021	HSPA5	-1.040491	0.003112	0.038810	Up in PC3 (FDR < 0.10)
O00468	AGRN	1.066182	0.003593	0.040089	Up in PC3M (FDR < 0.10)
Q12907	LMAN2	-1.699602	0.004943	0.045752	Up in PC3 (FDR < 0.10)
P02751	FN1	3.761799	0.004948	0.045752	Up in PC3M (FDR < 0.10)
Q99988	GDF15	-1.468005	0.004964	0.045752	Up in PC3 (FDR < 0.10)
O00622	CCN1	1.705974	0.005481	0.046480	Up in PC3M (FDR < 0.10)
P51648	ALDH3A2	-1.228481	0.006598	0.051302	Up in PC3 (FDR < 0.10)
P04439	HLA-A	1.031027	0.007163	0.051302	Up in PC3M (FDR < 0.10)
P24821	TNC	2.106766	0.009586	0.065559	Up in PC3M (FDR < 0.10)
Q15084	PDIA6	-1.020718	0.011280	0.068747	Up in PC3 (FDR < 0.10)
Q14697	GANAB	-1.112822	0.013495	0.075071	Up in PC3 (FDR < 0.10)
Q9P0M6	MACROH2A2	1.483157	0.013367	0.075071	Up in PC3M (FDR < 0.10)
Q9Y696	CLIC4	1.350660	0.015423	0.081743	Up in PC3M (FDR < 0.10)
P05556	ITGB1	1.387205	0.016146	0.083489	Up in PC3M (FDR < 0.10)
Q8TCT9	HM13	-1.768839	0.021176	0.089788	Up in PC3 (FDR < 0.10)

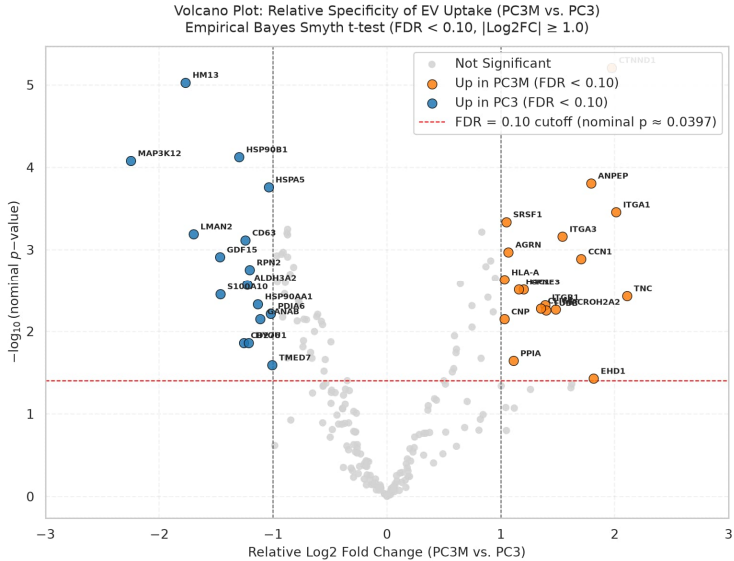
Empirical Bayes Moderated Differential Uptake Analysis

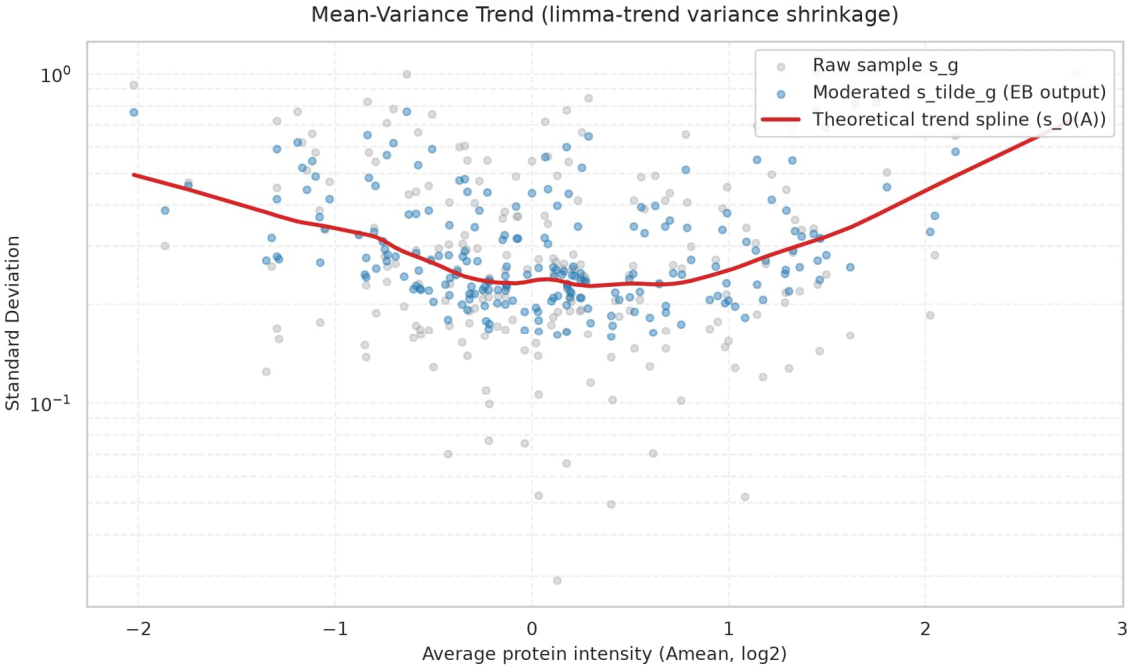
Since we do not have many samples (4+2), how about stabilizing dispersion with other peptides?

Empirical Bayes is being used to shrink sample variance and compute t-statistics

Multiple testing correction is applied using the Benjamini–Hochberg False Discovery Rate:

FDR < 0.10,
log_2(FC) >= 1.0

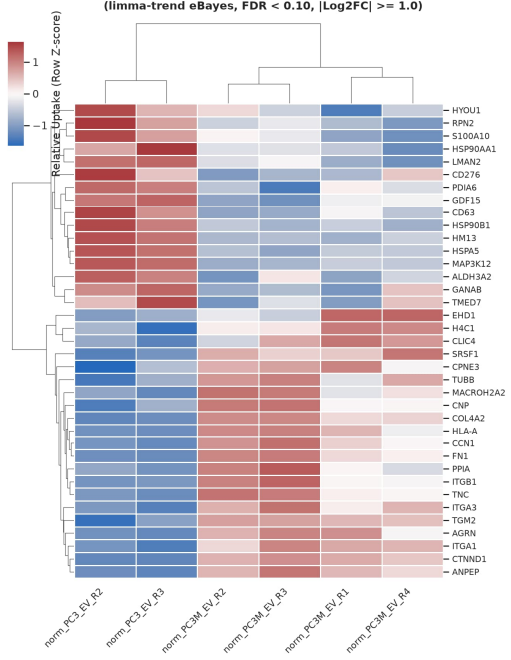



Mean-Variance Trend (limma-trend variance shrinkage)
Raw sample s_g
Moderated s_tilde_g (EB output)
Theoretical trend spline (s_0(A))
Standard Deviation
10^{0}
10^{-1}
−2
−1
0
1
2
3
Average protein intensity (Amean, log2)

Total proteins passing FDR < 0.10 and |Log2FC| >= 1.0: 37							
Up in PC3				Up in PC3M			
Gene	Log2FC	p_value	q_value	Gene	Log2FC	p_value	q_value
HM13	-1.8	0.00001	0.00077	CTNND1	2.0	0.00001	0.00077
HSP90B1	-1.3	0.00008	0.00294	TGM2	4.1	0.00001	0.00077
MAP3K12	-2.2	0.00008	0.00294	COL4A2	6.0	0.00005	0.00266
HSPA5	-1.0	0.00017	0.00461	ANPEP	1.8	0.00016	0.00461
LMAN2	-1.7	0.00065	0.00864	ITGA1	2.0	0.00035	0.00768
CD63	-1.2	0.00077	0.00903	FN1	3.8	0.00036	0.00768
GDF15	-1.5	0.00123	0.01089	SRSF1	1.0	0.00046	0.00864
RPN2	-1.2	0.00176	0.01287	ITGA3	1.5	0.00069	0.00864
ALDH3A2	-1.2	0.00272	0.01599	AGRN	1.1	0.00107	0.01089
S100A10	-1.5	0.00351	0.01862	CCN1	1.7	0.00130	0.01101
HSP90AA1	-1.1	0.00467	0.02198	HLA-A	1.0	0.00233	0.01541
PDIA6	-1.0	0.00609	0.02482	CPNE3	1.2	0.00309	0.01722
GANAB	-1.1	0.00708	0.02597	H4C1	1.2	0.00308	0.01722
HYOU1	-1.2	0.01376	0.04233	TNC	2.1	0.00373	0.01929
CD276	-1.3	0.01378	0.04233	ITGB1	1.4	0.00479	0.02202
TMED7	-1.0	0.02566	0.06885	CLIC4	1.4	0.00526	0.02323
				MACROH2A2	1.5	0.00544	0.02352
				TUBB	1.4	0.00560	0.02374
				CNP	1.0	0.00713	0.02597
				PPIA	1.1	0.02276	0.06185
				EHD1	1.8	0.03729	0.09524

Hierarchical Clustering

To evaluate whether the identified differentially abundant proteins capture coherent multivariate differences between the conditions without model supervision, we performed unsupervised hierarchical clustering on the significant hits



Thanks for your attention!
All the materials can be found in my GitHub:



https://github.com/Dudkanist/ev-uptake-proteomics

My previous attempts to apply ML (declined, as it is difficult to interpret the results of fitting models to 6 samples when there are a thousand parameters involved) and Pathway analysis (it was preliminary attempt just to familiarize myself with this tools)

Supervised Machine Learning: Random Forest Feature Importance

15,000 Decision Trees (depth = 1) work as binary test:

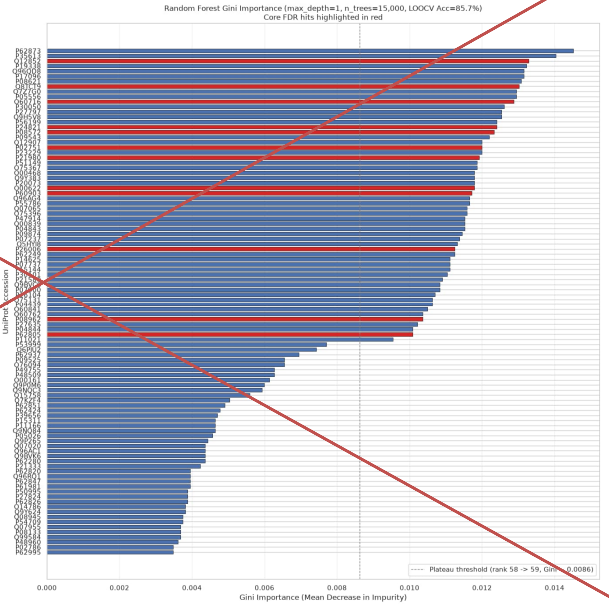
1. Subsampling: For each tree, a random subset of 12 proteins $(\sqrt{159})$ is selected.

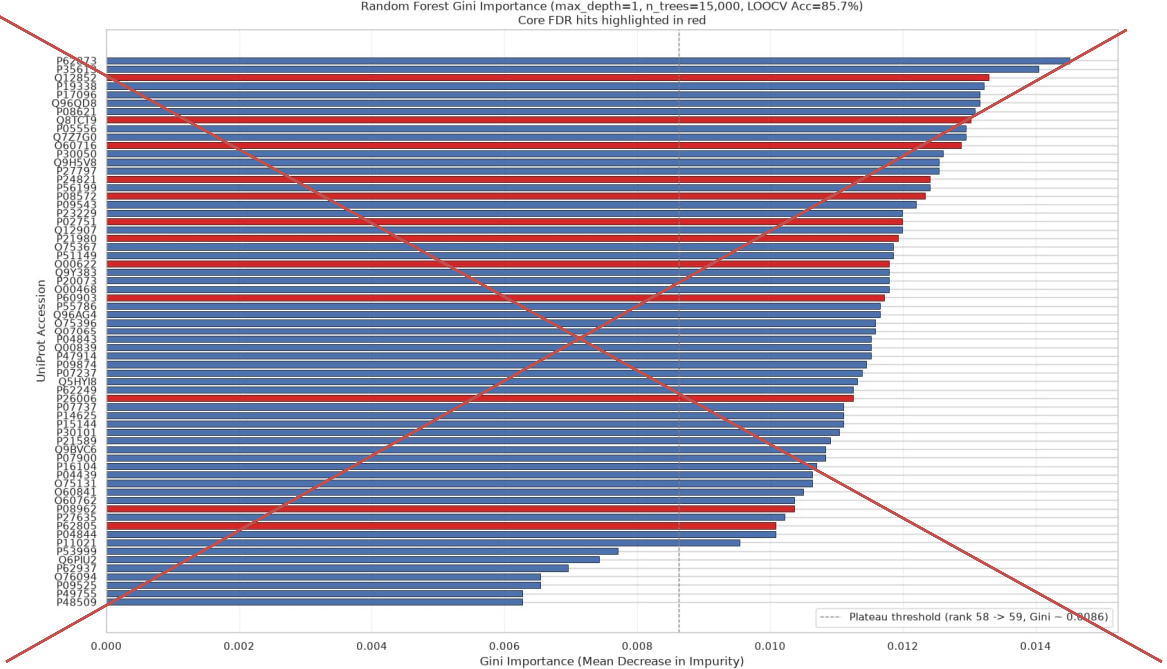
2. Single Decision Split: The tree evaluates only one threshold on the best-performing protein to separate the 7 samples into PC3 and PC3M, calculating the drop in Gini impurity.

3. (Monte Carlo) Convergence: Repeating this 15,000 times eliminates sampling noise, giving every protein equal opportunity to compete and stabilizing the feature rankings.

4. Result: The top-ranked proteins are those that consistently achieved the cleanest binary split across thousands of random iterations.

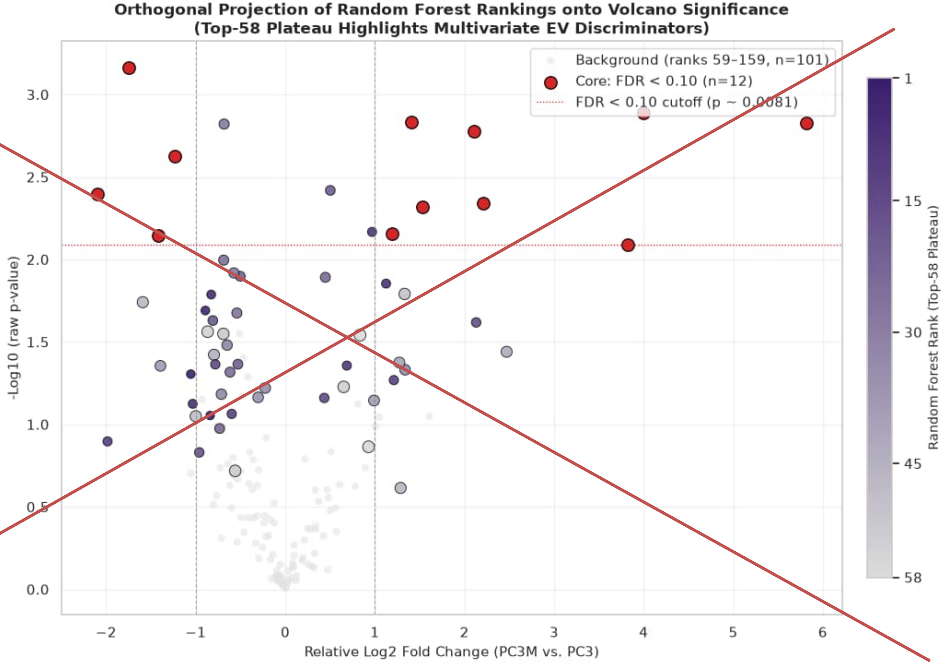
The sharp plateau of 58 proteins identifies the complete biological pool capable of linearly separating PC3 from PC3M with zero classification error. Our 12 FDR-significant hits represent the high-amplitude core of this pool.



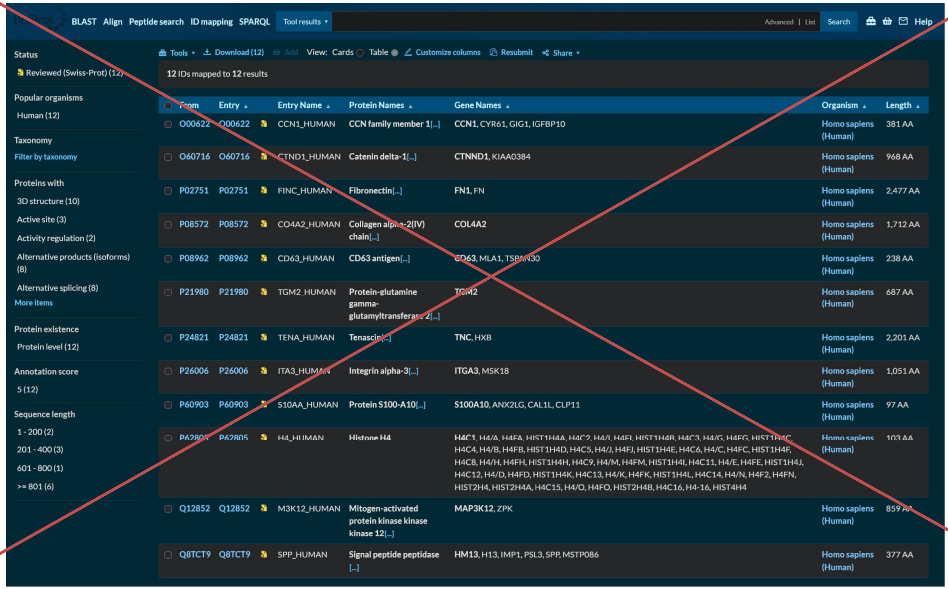

Random Forest Gini Importance (max_depth=1, n_trees=15,000, LOOCV Acc=85.7%)
Core FDR hits highlighted in red
UniProt Accession
P62873
P35613
Q12834
P19338
P17096
Q96QD8
P08621
Q8TCT0
P05556
Q7Z7G0
Q60716
P30050
Q8HSV8
P27797
P24821
P56199
P08572
P09543
P23229
P02751
Q12907
P21980
Q75367
P51149
O00622
Q9Y383
P20073
O00468
P60903
P55786
Q96AC4
Q75390
Q07065
P04843
Q00839
P47914
P09874
P07237
Q5HYI8
P62249
P26006
P07737
P14625
P15144
P30101
P21589
Q9BVC6
P07900
P16104
P04439
Q75131
Q60841
Q60762
P08962
P27635
P62805
P04844
P11021
P53999
Q6PIU2
P62277
Q76094
P09525
P46755
P48509
Plateau threshold (rank 58 -> 59, Gini ~ 0.0086)
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
Gini Importance (Mean Decrease in Impurity)

To examine the agreement between univariate hypothesis testing and multivariate machine learning, we project Random Forest importance ranks onto the Volcano plot.

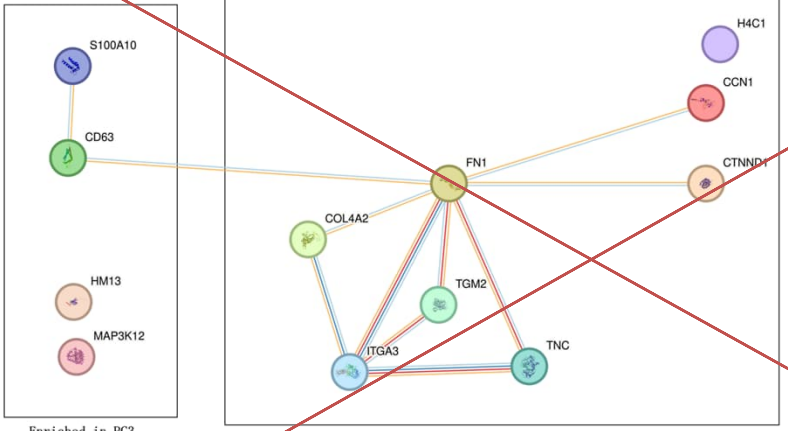
A continuous color and point-size gradient highlights proteins within the top-58 importance plateau that fall just below stringent FDR thresholds due to degrees-of-freedom penalties.



Uniprot: looking at significant proteins



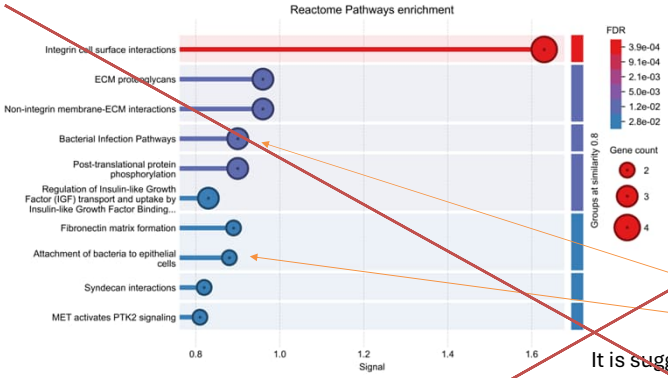
STRING PPI network analysis of 12 differentially quantified proteins (FDR < 0.10, Log(FC) > 1.0)



https://version-12-5.string-db.org/cgi/network?networkId=bVoQO0r6vsvh

- Nodes represent query proteins
- Edges indicate confidence-supported interactions (from experimental assays, databases, and co-expression)

The interactome shows significant functional clustering *(PPI enrichment p ~ 10^-7, 11 edges vs. 1 expected for any 11 random given proteins)* centered on FN1 and its matrix-integrin binding partners



It is suggested that bacteria employ similar mechanisms to attach themselves to human fibronectin (FN1) and collagen by binding to integrins on the surface of the host's cells.

The mechanism by which nanoparticles and vesicles adhere to a recipient cell prior to endocytosis might be similar to the mechanism by which bacteria attach to epithelium via a matrix adhesive.

Reactome Pathways enrichment supports the hypothesis:
The uptake(?) and contact of vesicles in metastatic cells are linked to surface integrins.